

E-Cart Viva Presentation Script & Defense Guide

Simple, Concise & Easy-to-Memorize Guide

1. Quick 30-Second Viva Pitch (What to Say)

"Good morning / afternoon, Respected Examiners."

I am presenting E-Cart, an AI-powered E-Commerce platform with an Intelligent RAG Shopping Advisor.

Unlike ordinary chatbots that make up fake answers, our system uses a 3-Stage RAG Pipeline:

- Retrieval:** Searches our real store database for exact matching items.
- Augmentation:** Prepares real product specs and prices in Indian Rupees (₹).
- Generation:** Generates answers strictly from store data, refusing any off-topic questions.

It supports 28+ real tech products, a 5-Item Gaming Setup Builder, and a Full-Page Product View with verified buyer reviews."

2. System Architecture Diagram

Below is the 3-Stage RAG Pipeline flow:



3. Top 5 Simple Viva Questions & Answers

Q1: What is RAG?

Answer: RAG stands for **Retrieval-Augmented Generation**. It searches our real store catalog first before giving answers, so the AI never guesses or makes up fake products.

Q2: How do you stop fake or hallucinated answers?

Answer: We set a similarity threshold (0.25). If a question is off-topic (like sports or cars), the AI immediately says: "I can only answer about products in our E-Cart catalog."

Q3: Which embedding model is used for vector search?

Answer: We use [sentence-transformers/all-MiniLM-L6-v2](#). It converts text into 384-dimensional vectors for fast, accurate search.

Q4: How does the Gaming Setup Builder work?

Answer: When a user asks for a gaming setup, it automatically selects 5 matching items (PC, OLED Monitor, Mouse, Keyboard, Headset) and calculates total price in ■.

Q5: What backend technologies did you use?

Answer: Python, FastAPI for REST APIs, SentenceTransformers for vector search, and Gemini AI for grounded responses.

■ ■ 4. 4-Step Live Demo Script (What to Click)

1. **Show Home Page:** Point out category tabs (Smartphones, Laptops, Cameras, Smartwatches).
2. **Open Full-Page View:** Click any product to show high-res image, spec table, and verified buyer reviews.
3. **Test Gaming Setup:** Open chatbot and click ■ [Build Gaming Setup](#) to show 5-item bundle with 1-click cart buttons.
4. **Test Category & Guardrail:**
5. Type ["camera under 50 k"](#) → Show real cameras (GoPro HERO 12 & Sony ZV-E10), zero phones.
6. Type ["who won the world cup?"](#) → Show AI refusal guardrail message.